

AUTONOMOUS MARS ROVER (ARES-1)

Track 2: Autonomous Mars Rover (Unit 3 - Propositional Logic Agent)

Propositional Logic Inference (PL-Resolution) • Dynamic KB Console Logs • 100%

Solvable

Course: Artificial Intelligence (Unit 3 Logic Agents)

Group Name / ID: Group 6

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GitHub Repository: https://github.com/darsshann011/Ai_Rover

1. PEAS FRAMEWORK MATRIX

Component	Specification Details & Implementation Invariants (Scenario & Deliverables)
Performance Measure (P)	100% Goal Reachability: Successfully navigates from Landing Pad (0,0) to Extraction Beacon (N-1, N-1). Zero Safety Violations: Never enters Hazard (Acid) or Radiation (Gas) cell (0 violations verified across tests). Inference Efficiency: Real-time live dynamic KB logging per step; minimal resolution overhead via Unit Propagation.
Environment (E)	Martian 2D Grid: $N \times N$ discrete planetary surface containing <i>SAFE</i> terrain, <i>HAZARDS</i>, and <i>RADIATION</i> zones. Properties: Partially observable, deterministic state transitions, static layout, discrete time steps. Solvability Guarantee: Validated via BFS pathfinding; automatic corridor injection guarantees 100% solvable maps.
Actuators (A)	Holonomic 4-Dir Drive: Movement commands $MOVE(x, y)$ to adjacent orthogonal coordinates $\{(x\pm 1, y), (x, y\pm 1)\}$. Heading Rotation & Tire Tracks: Directional facing update vector with persistent trajectory logging. Global Safe Backtracker: Automated BFS sequencer returning rover along confirmed safe visited cells when dead-ended.
Sensors (S)	Local Hazard Detector: Senses <i>HazardSignal</i> at Manhattan distance $d \leq 1$ (adjacent cells + current cell). Radiation Geiger / Spectrometer: Senses <i>RadiationSignal</i> at Manhattan distance $d \leq 2$ (extended warning range). Odometry & Signal Classifier: Senses <i>NoSignal</i> on clear cells; odometry yields exact current position (x, y).

2. CORE ALGORITHMIC FORMULATION & PROPOSITIONAL LOGIC (KB-AGENT)

State Space (S):	$s = (x, y, KB, Visited, Frontier)$ where $(x, y) \in [0, N-1]^2$ and KB is the active CNF propositional sentence set.	Initial State (s$\blacksquare$):	$s\blacksquare = (0, 0, KB\blacksquare, \{(0,0)\}, \emptyset)$ with axioms $\{Safe_ (0,0), \neg HazardSignal_ (0,0), \neg RadiationSignal_ (0,0)\}$.
Goal Test & Cost:	$GoalTest(s) \equiv ((x, y) == (N-1, N-1))$. Step cost $c(s,a,s') = 1$; cost into hazard/radiation $c = +\infty$ (strictly forbidden).	Entailment Check:	Move to (nx, ny) iff $KB \blacksquare Safe_ (nx,ny)$ proven by refutation: $KB \cup \{\neg Safe_ (nx,ny)\} \blacksquare_{PL-Res} \emptyset$.
Propositional Logic KB Rules (CNF Clauses)		Goal-Directed Heuristic & Dynamic Logging Delivery	
Hazard Avoidance Rule: $Perceive\ HazardSignal(x,y) \rightarrow \neg MoveForward / Blocked(x,y)$ CNF: $\{ \neg HazardSignal_ (x,y), Blocked_ (x,y) \}$ Radiation Avoidance Rule: $RadiationSignal(x,y) \rightarrow Blocked(x,y)$ CNF: $\{ \neg RadiationSignal_ (x,y), Blocked_ (x,y) \}$ Safety Equivalence: $(\neg Hazard \wedge \neg Radiation) \rightarrow Safe_ (x,y)$ CNF: $\{ HazardSignal_ (x,y), RadiationSignal_ (x,y), Safe_ (x,y) \}$ Blocked Invariant: $Blocked_ (x,y) \rightarrow \neg Safe_ (x,y)$ CNF: $\{ \neg Blocked_ (x,y), \neg Safe_ (x,y) \}$ Unit Propagation: Eager unit clause inference executed on every TELL.		Goal-Directed A* Heuristic: $score(n) = (n_x - G_x + n_y - G_y) \cdot \lambda \cdot UnexploredNeighbors(n)$ Prioritizes KB-inferred safe neighbors closest to Goal (G_x, G_y). Global Safe Backtracking Policy: When local path is blocked, agent locates closest unexpanded safe frontier v : $cost(v) = dist_BFS(pos, v) + dist_Manhattan(v, Goal)$ Dynamic Live Console Logging (Deliverable): Synchronized per-step output showing: <i>PERCEIVE</i> signals $\rightarrow$ <i>TELL</i> (clause additions & unit propagation) $\rightarrow$ <i>ASK</i> resolution refutations $\rightarrow$ <i>DECIDE</i> action.	

3. COMPLEXITY ANALYSIS: THEORETICAL (O-NOTATION) VS. OBSERVED BENCHMARKS

Module / Operation	Theoretical Time Complexity	Theoretical Space Complexity	Observed Benchmark Performance (6x6 Grid)
TELL & Unit Propagation	$O(C \cdot L)$ where C = clause count, L = literals per clause. $O(1)$ amortized per percept.	$O(N^2)$ storing derived unit facts in dictionary <i>facts</i> .	< 0.15 ms per step. Eagerly derives 100% of direct safe/hazard facts without invoking full resolution.
PL-RESOLUTION (ASK)	$O(2^V)$ worst-case over V proposition symbols; bounded by $MAX_STEPS = 5000$.	$O(C^2)$ resolvent working clause set storage.	< 1.10 ms per query. Direct fact lookup satisfies ~94% of queries; 0 step-limit timeouts encountered.
Frontier BFS Backtracking	$O(V + E) = O(N^2)$ where $V \leq N^2$ visited safe nodes and $E \leq 4N^2$ edges.	$O(N^2)$ for BFS queue and <i>visited</i> coordinates set.	< 0.35 ms path generation. Flawless backtracking execution over known visited safe routes.
End-to-End Mission Loop	$O(K \cdot N^2)$ total mission time where $K \leq 300$ max exploration steps.	$O(N^2)$ total footprint (KB clauses + trail history).	100% Goal Reachability (50/50 test pass); 0.00% safety violations ; avg 11-18 steps to extraction.